Python Activity-Based Learning
28. Repeated Transaction Detection

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

A shop records transaction numbers in a list. Write a program to find and display transaction numbers that appear more than once.

2. Objectives

• Store transaction numbers in a list.

• Check whether a number has appeared before.

• Display duplicate transaction numbers.

3. Assumptions

• Transaction numbers are entered as strings.

• The same duplicate number is displayed only once.

4. Algorithm (Step by Step)

1. Start.

2. Read the number of transactions.

3. Read all transaction numbers into a list.

4. Create an empty list for duplicates.

5. Check each transaction number.

6. If it is already in the seen list and not in duplicates, add it to duplicates.

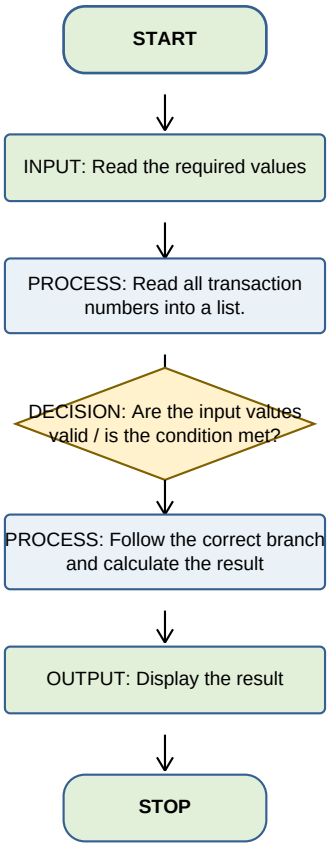
7. Otherwise, add it to the seen list.

8. Display the duplicates or state that none were found.

9. Stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
INPUT n
transactions = empty list
REPEAT n TIMES: INPUT id; ADD id to transactions
seen = empty list; duplicates = empty list
FOR each id in transactions
    IF id in seen AND id not in duplicates THEN ADD id to duplicates
    ELSE ADD id to seen
PRINT duplicates
STOP
```

7. Python Source Code

```
n = int(input("How many transactions? "))
transactions = []

for i in range(n):
    value = input("Enter transaction ID: ")
    transactions.append(value)

seen = []
duplicates = []

for value in transactions:
    if value in seen:
        if value not in duplicates:
            duplicates.append(value)
    else:
        seen.append(value)

if duplicates:
    print("Repeated transaction IDs:", duplicates)
else:
    print("No repeated transactions")
```

8. Test Cases

Input / Test	Expected result	What it checks
A1, B2, A1, C3, B2	['A1', 'B2']	Two IDs are repeated
X1, X2, X3	No repeated transactions	All IDs are unique
P1, P1	['P1']	One ID is repeated

9. Sample Output

```
How many transactions? 4
Enter transaction ID: A1
Enter transaction ID: B2
Enter transaction ID: A1
Enter transaction ID: C3
Repeated transaction IDs: ['A1']
```

10. Conclusion

This program uses lists and loops to identify repeated transaction numbers.